

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
 2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
 3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
 4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
 5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.
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Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

ChatGPT Answer

L1, L2, and L3 are different levels of cache memory used to reduce the time required by the processor to access data.

Cache	Latency	Bandwidth	Throughput	Main Feature
L1	Very Low	Very High	Very High	Smallest and fastest
L2	Low	High	High	Larger than L1
L3	Higher than L1/L2	High	High	Larger and usually shared

L1 cache is located closest to the CPU core and provides the fastest access. It stores frequently used instructions and data. **L2 cache** has greater capacity but slightly higher latency. **L3 cache** is larger and is often shared between processor cores.

A higher cache hit rate means the processor can obtain data without accessing slower main memory. Therefore, L1 has the greatest effect on reducing latency, while L2 and L3 help reduce the number of DRAM accesses. A well-designed cache hierarchy improves processor throughput and overall performance.

Gemini AI Answer

L1, L2, and L3 caches form a hierarchy between the processor and main memory. Their main difference is the balance between speed and capacity.

- **L1 Cache:** Fastest cache with the lowest access latency. It has limited capacity and very high bandwidth.
- **L2 Cache:** Provides more storage than L1 while maintaining relatively fast access. Its latency is higher than L1.
- **L3 Cache:** Provides significantly greater capacity and is generally shared among multiple cores. It is slower than L1 and L2 but much faster than DRAM.

The cache hierarchy improves processor performance by keeping frequently accessed information close to the CPU. When data is found in L1, the processor gets the lowest latency. If there is an L1 miss, L2 and then L3 are checked before accessing DRAM. Thus, increasing cache hit rates improves effective memory access time and processor throughput.

Claude AI Answer

Cache levels differ primarily in size, speed, and distance from the processor.

L1 cache provides the lowest latency and highest effective access speed but has the smallest capacity. **L2 cache** offers a compromise between capacity and speed. **L3 cache** is considerably larger and is commonly shared by processor cores, but its access latency is higher.

The hierarchy works by placing the most frequently accessed information in the fastest cache. A cache hit allows the CPU to avoid a slower memory access, while a cache miss requires searching the next level. Therefore, efficient cache management and high hit rates reduce average memory access time.

From a performance perspective, L1 has the strongest influence on latency, while L2 and L3 are important for reducing DRAM accesses and maintaining processor throughput.

Comparison

Aspect	ChatGPT	Gemini	Claude
Explanation	Balanced	More structured	More conceptual
Technical depth	Moderate	Moderate	High
Readability	High	High	High
Main focus	Performance	Cache hierarchy	Memory access behavior

2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

ChatGPT Answer

SRAM and DRAM are both volatile memories but have different performance characteristics.

Feature	SRAM	DRAM
Latency	Very Low	Higher
Bandwidth	High	High
Throughput	Very High	High
Density	Low	High
Typical Use	Cache	Main Memory

SRAM uses flip-flop-based storage and does not require periodic refreshing. It provides very low latency, making it suitable for L1, L2 and L3 cache. However, SRAM requires more transistors per bit, making it expensive and less dense.

DRAM stores data using capacitors and requires periodic refreshing. It has higher latency than SRAM but provides much greater capacity at a lower cost. Therefore, it is commonly used as the main memory.

Thus, SRAM is preferred when low latency is important, while DRAM is preferred when large capacity and good bandwidth are required.

Gemini AI Answer

SRAM and DRAM serve different roles in a computer's memory hierarchy.

SRAM offers faster access because it does not need refresh operations. Its low latency and high bandwidth make it suitable for processor cache. However, its high cost and lower density limit the amount that can be used.

DRAM has higher access latency because the stored charge must be maintained through refresh operations. However, it offers high density and large capacity, making it suitable for main memory.

The combination of SRAM and DRAM provides an effective balance: SRAM supplies fast data access near the processor, while DRAM provides the larger memory space required by applications.

Claude AI Answer

The major difference between SRAM and DRAM is the underlying storage method, which directly affects their performance.

SRAM uses bistable circuits to retain information as long as power is supplied. This allows very fast access and makes SRAM appropriate for cache memory. DRAM uses capacitors that gradually lose their charge, requiring periodic refresh cycles. This increases latency but allows much higher memory density.

In cache design, SRAM is valuable because cache accesses occur frequently and directly affect CPU performance. In main memory design, DRAM is more practical because its higher density allows large capacities at a lower cost.

Therefore, SRAM prioritizes **latency**, whereas DRAM provides a better balance of **capacity, bandwidth and cost**.

Comparison

Aspect	ChatGPT	Gemini	Claude
SRAM explanation	Direct	Simple	Detailed
DRAM explanation	Balanced	Easy to understand	Technical
Cache/Main memory connection	Clear	Clear	Strong
Overall	Good	Very readable	Most analytical

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

ChatGPT Answer

Cache memory and virtual memory with paging are used for different purposes.

Cache memory is a small and fast memory located close to the processor. It stores frequently accessed data and instructions. Cache access has very low latency, and cache hits improve system throughput.

Virtual memory with paging allows the operating system to use secondary storage when physical RAM is insufficient. Memory is divided into pages and physical memory into frames. If a required page is not in RAM, a page fault occurs and it must be loaded from storage. This process has much higher latency than cache access.

Feature	Cache Memory	Virtual Memory/Paging
Purpose	Speed up access	Extend memory capacity
Latency	Very Low	High during page fault
Storage	Small	Large
Effect on throughput	Increases	Can decrease if paging is frequent

Therefore, cache primarily improves **speed and throughput**, while virtual memory provides **larger usable memory capacity**.

Gemini AI Answer

Cache and virtual memory solve different memory problems.

Cache memory improves processor performance by keeping frequently used data close to the CPU. Its access time is extremely small compared with RAM or storage.

Virtual memory provides an abstraction of a large memory space. With paging, data is transferred between DRAM and secondary storage in fixed-size pages. A page fault occurs when the required page is absent from physical memory. Since storage access is much slower, excessive paging can reduce system throughput.

A good system therefore tries to maximize cache hits and minimize page faults. Cache is mainly a performance mechanism, while paging is mainly a capacity and memory-management mechanism.

Claude AI Answer

Cache memory and virtual memory operate at different levels of the memory hierarchy.

Cache memory reduces **CPU access latency** by storing recently or frequently used data in fast memory. A cache hit allows immediate access with minimal delay.

Virtual memory with paging manages the relationship between virtual address space and physical memory. When a page is absent from RAM, a page fault causes the operating system to retrieve it from secondary storage. This introduces a very large latency compared with cache access.

Consequently, cache memory generally improves throughput by reducing CPU stalls, whereas excessive paging can severely reduce throughput because processors may spend significant time waiting for data.

Comparison

Aspect	ChatGPT	Gemini	Claude
Main focus	Direct comparison	Conceptual difference	System-level analysis
Latency explanation	Clear	Clear	Detailed
Throughput analysis	Good	Good	Strong
Best feature	Simple table	Easy explanation	Technical reasoning

4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

ChatGPT Answer

NAND Flash and DRAM have very different characteristics and are used at different levels of the memory hierarchy.

Feature	DRAM	NAND Flash
Latency	Low	High
Bandwidth	High	Moderate to High
Volatility	Volatile	Non-volatile
Capacity	High	Very High
Main Use	Main Memory	Secondary Storage

DRAM provides fast random access and high bandwidth, making it suitable for main memory. However, it is volatile and loses data when power is removed.

NAND Flash is non-volatile and can provide large storage capacity. It has higher access latency than DRAM, especially for random accesses. It is therefore normally placed below DRAM in the hierarchy and is used in SSDs and other storage devices.

A suitable hierarchy is:

CPU → Cache → DRAM → NAND Flash

This arrangement provides fast access through DRAM while using NAND Flash for large-capacity permanent storage.

Gemini AI Answer

DRAM and NAND Flash differ mainly in speed, storage capacity and data retention.

DRAM has lower latency and high bandwidth, which makes it suitable for active programs and frequently accessed data. It is volatile, so data must be maintained while power is available.

NAND Flash has much higher capacity and retains information without power. However, its access latency is significantly higher than DRAM, particularly for random data access.

In a hierarchical system, DRAM should be placed closer to the processor, while NAND Flash should be used for long-term storage. Data can be moved from NAND Flash into DRAM when required for processing.

Claude AI Answer DRAM and NAND Flash occupy different positions in the memory hierarchy because their performance characteristics are fundamentally different.

DRAM provides relatively low-latency random access and high memory bandwidth. These characteristics make it appropriate for holding the active working set of an application.

NAND Flash provides non-volatile storage with much higher density, but its latency is considerably higher than DRAM. It is therefore unsuitable for replacing DRAM as the primary working memory in latency-sensitive applications.

The appropriate hierarchy places **DRAM above NAND Flash**, allowing frequently accessed data to remain in fast memory while NAND Flash provides large, persistent storage.

Comparison

Aspect	ChatGPT	Gemini	Claude
Technical detail	Moderate	Moderate	High
Hierarchy explanation	Very clear	Clear	Strong
DRAM analysis	Good	Simple	Detailed
NAND analysis	Good	Easy	Analytical
Best feature	Practical hierarchy	Simplicity	Technical depth

5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

ChatGPT Answer

MRAM and RRAM are non-volatile memory technologies being considered for high-performance and low-power applications.

Feature	MRAM	RRAM
Access Time	Low	Low
Throughput	High	High
Energy Efficiency	High	High
Non-volatile	Yes	Yes
Main Advantage	Fast and reliable access	High density and low power potential

MRAM stores information using magnetic states. It provides fast access, non-volatility and good endurance. It can be used in embedded systems and as a possible alternative to some conventional memory technologies.

RRAM stores information by changing the resistance of a material. It can provide low-power operation and high density. Its simple structure makes it attractive for future memory systems.

Both technologies can reduce standby power because they retain data without continuous power. Their exact performance depends on the specific implementation and device technology.

Gemini AI Answer

MRAM and RRAM are both non-volatile memories, but they use different methods to store information.

MRAM uses magnetic states to represent data. It provides fast read/write operations, good endurance and low standby power because data is retained without continuous power.

RRAM uses changes in electrical resistance to store data. It has the potential for high density, low operating energy and fast switching.

Both technologies are useful for applications where low power and fast access are required. MRAM is particularly attractive when endurance and speed are important, while RRAM is promising for high-density and low-power applications.

Claude AI Answer

MRAM and RRAM are emerging non-volatile memory technologies designed to reduce some limitations of conventional volatile memory.

MRAM stores data using magnetic orientation, allowing it to retain information without power. It offers fast access, high endurance and relatively low standby energy.

RRAM uses changes in resistance within a memory cell to represent different data states. Its simple cell structure and potential for high density make it attractive for future memory architectures. It can also offer low switching energy.

In terms of throughput, both technologies can provide high performance, but actual results depend heavily on device architecture, read/write operations and implementation. MRAM is generally attractive for applications requiring high endurance and fast access, while RRAM has strong potential for dense, energy-efficient memory.

Comparison

Aspect	ChatGPT	Gemini	Claude
MRAM explanation	Balanced	Simple	Detailed
RRAM explanation	Balanced	Easy	Technical
Energy efficiency	Good	Clear	Strong
Throughput discussion	Good	Moderate	Most detailed
Best feature	Overall balance	Simple language	Technical analysis

Overall Comparison of the Three AI Tools

Criteria	ChatGPT	Gemini AI	Claude AI
Clarity	Excellent	Excellent	Very Good
Technical detail	Very Good	Good	Excellent
Simple explanation	Excellent	Excellent	Good
Tables/comparison	Excellent	Good	Good
Analytical approach	Very Good	Good	Excellent
Student-friendly	Excellent	Excellent	Very Good
Overall usefulness	Very High	High	Very High